🔐 CyberLock – ESP32-CAM Based Smart Door Lock System

CyberLock is an IoT-enabled smart door lock system built using the **ESP32-CAM module**, designed to provide secure, automated, and keyless access control through real-time facial recognition.

The system authenticates authorized users through facial recognition and automatically controls a 12V solenoid door lock, offering a modern alternative to traditional key-based security systems.

🚀 Project Overview

CyberLock integrates embedded systems, IoT communication, computer vision, and electronic hardware to create an intelligent access control solution.

The ESP32-CAM acts as the main controller, handling camera operations, facial recognition, and a local web-based interface for managing access.

✨ Key Features

- **Face Recognition:** Authenticates registered users for secure access.
- **Live Video Streaming:** Provides a real-time camera feed through a web interface.
- **Wi-Fi Access Point:** Creates a dedicated Wi-Fi network named `CyberLog`.
- **Face Enrollment:** Allows administrators to register new users.
- **Automated Door Locking:** Controls a 12V solenoid lock through a relay module.
- **Automatic Relocking:** Secures the door again after a predefined unlocking delay.
- **Hardware Protection:** Uses a 1N4007 flyback diode to protect the circuit from inductive voltage spikes.

🛠 Hardware Components

Component	Purpose
ESP32-CAM	Main controller, camera, and facial recognition
1-Channel Relay Module	Controls the solenoid lock
12V Solenoid Door Lock	Electronic locking mechanism
1N4007 Diode	Flyback protection
Power Supply	Provides the required operating voltages
Jumper Wires	Hardware interconnections

💻 Software & Technologies

- **Programming Language:** C++
- **Development Environment:** Arduino IDE
- **Microcontroller:** ESP32-CAM
- **Communication:** Wi-Fi
- **Interface:** Local Web Server
- **Technology:** IoT, Embedded Systems, Facial Recognition

⚙ Working Principle

1. **System Initialization:** The ESP32-CAM initializes the camera, Wi-Fi, and web server.
2. **Wi-Fi Connection:** The device broadcasts a Wi-Fi access point named `CyberLog`.
3. **User Access:** An administrator connects to the network and opens the local web interface.
4. **Face Enrollment:** New authorized users can be registered through the interface.
5. **Authentication:** The camera captures the user's face and checks it against registered faces.
6. **Door Unlocking:** When an authorized face is recognized, the ESP32-CAM triggers the relay using an active-low signal.
7. **Automatic Relocking:** The solenoid lock is de-energized after the configured delay, securing the door again.

🔌 Hardware Design & Protection

The locking mechanism uses a relay module configured with its Normally Open (NO) contact to control the 12V solenoid lock.

An important hardware protection feature is the **1N4007 flyback diode**, connected in parallel with the solenoid coil with reverse polarity during normal operation.

The diode suppresses the inductive voltage spike generated when the solenoid is switched off, helping protect the relay contacts and electronic components.

📂 Repository Structure

```
CyberLock/
│
├── firmware/
│   └── CyberLock.ino
│
├── hardware/
│   ├── circuit_diagram/
│   └── components_list.md
│
```

```
├── images/
│   └── project_setup.jpg
│
└── README.md
```

🎯 Project Objectives

- Develop a smart and automated access control system.
- Implement facial recognition using the ESP32-CAM.
- Integrate IoT-based monitoring and administration.
- Interface a microcontroller with an electronic locking mechanism.
- Apply hardware protection techniques in an embedded system.

🔮 Future Improvements

- Mobile application integration.
- Remote access monitoring.
- Access logs and authentication history.
- Multiple administrator accounts.
- Additional security layers and authentication methods.

🕵️‍♂️💻 Project Focus

CyberLock demonstrates the practical application of **Internet of Things (IoT), embedded systems, C++ programming, facial recognition, and electronic hardware interfacing** in modern smart security solutions.

Project Name: ESP32-CAM-Face-Recognition-Lock
Programming Language: C++
Development Environment: Arduino IDE
Category: IoT | Embedded Systems | Smart Security